Three Results in Distributed Discovery

Roles, Disclosure, and Source Dependence

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Abstract

Collective search depends on more than the quality of a shared ranking. It also depends on who is assigned to test what, which strategic outcome is selected, and whether nominally separate reports arise from distinct sources. We synthesize three finite results around those margins. First, ex-ante role policies can improve discovery without communication: at $(M, N, p) = (3, 2, 2/5)$ a hybrid policy attains $7/10$, strictly above direct clue-following $16/25$, and a restricted family has exact territorial–hybrid–direct thresholds. Second, in a complete deterministic disclosure fixture, one Blackwell refinement lowers discovery under the declared anonymous-symmetric selection from $5/9$ to $171/308$, even though every pure equilibrium and the planner improve. An exact six-rule catalogue shows that this known reversal survives only anonymous-symmetric selection among the audited alternatives. Third, among 51 nonisomorphic latent-source graphs with homogeneous accuracy, complete pairwise report moments yield a bounded null. In a registered 839-network colored extension, however, two networks have identical complete first/pairwise moments but discovery $3/4$ and $2/3$; average agreement is also insufficient. We connect these results through a common information–action framework and preserve their distinct evidence boundaries. For the canonical private-team fixture, an alignment-preserving exact relaxation also meets the direct-policy lower bound, proving direct clue-following globally optimal without communication. The paper makes no claim of a new field or universal law.

Keywords: collective search, team decision, information design, source dependence, equilibrium selection, distributed discovery.

Evidence note: All displayed numerics and generated figures are derived from checksum-validated immutable runs. Claim IDs refer to the public repository ledger; exactness and independence are claim-specific.

1 Introduction

Many organizations use the language of "the best answer" even when their operational task is to allocate several attempts. A research group chooses a portfolio of experiments, a venture fund chooses investments, and a multi-agent system chooses investigations or tool calls. In each case evidence ranks candidates, but discovery depends on the union of actions rather than the quality of a single recommendation. The Shared Discovery Paradox isolates this difference: pooling improves the leading recommendation, yet a one-answer convention can compress several actions into repetition and reduce discovery [Nakajima, 2026].

The broader program calls this conversion from dispersed evidence into a finite action portfolio *distributed discovery*. The name is a working definition, not a novelty or ownership claim. The phrase has other uses, and the component questions belong to established literatures: Blackwell comparison and persuasion for information, team theory for private common-payoff decisions, organizational and scientific search for division of effort, and coverage games for strategic action allocation [Blackwell, 1953, Marschak and Radner, 1972, Kamenica and Gentzkow, 2011, March, 1991, Kitcher, 1990, Vetta, 2002]. The contribution here is to place three finite results in one explicit architecture while retaining their separate scopes.

The first result concerns *roles*. Agents commit to different mappings from private clues to actions before seeing those clues and act without communication. Direct clue-following is feasible but need not be optimal. A constant territory can reserve a candidate while another agent remains responsive to evidence. This mechanism is allocative: no new signal is created, and no report is pooled at action time.

The second result concerns *selection*. A public disclosure changes posterior games played by autonomous searchers. More informative disclosure expands what a planner can do, but a named equilibrium-selection rule can move toward a more concentrated outcome. The finite reversal below is therefore related in spirit to informational Braess effects, not to a failure of frontier monotonicity [Acemoglu et al., 2018]. Its central qualifier is that all pure equilibria improve for the same witness, and an exact catalogue shows that five alternative rules do not reproduce that witness.

The third result concerns *representation*. Correlated reports may come from latent source networks. One scalar summary of agreement can hide discovery- relevant structure. A homogeneous bounded class records where the complete pairwise matrix did not fail; exact accuracy colors then produce nonisomorphic networks with the same full pairwise object and different discovery. The positive counterexamples and preserved negative search identify the perturbation that breaks the earlier null.

These mechanisms should not be collapsed into a slogan that "diversity is good" or "more information is bad." Roles change the feasible private policy class; disclosure changes a posterior game and its selected equilibrium; source networks change the joint evidence law. Our unifying statement is narrower: collective discovery depends not only on what agents know, but on how roles are assigned, how strategic outcomes are selected, and how source dependence is represented.

The remainder defines the common accounting and evidence discipline, presents the three results, draws their limited joint implications, and states the unresolved questions. The canonical upstream paper remains the concise source for the atomic paradox and its strategic/scaling results. The Foundations note remains the source for the general vocabulary. This paper is a results synthesis, not a replacement for either.

2 Framework and evidence discipline

2.1 Information and action as separate objects

An atomic discovery problem has a target $\theta \in \{1, \ldots, M\}$ and N actions. If S is the set of distinct tested labels, discovery is $\mathbf{1}\{\theta \in S\}$. An information architecture specifies which evidence each decision maker observes. A protocol maps that information to a feasible action vector. With budget B, information architecture I, and feasible protocol class $\mathcal{P}_B(I)$, define

$$G_B(\Pi; I) = \mathbb{P}_\Pi(\theta \in S), \qquad V_B(I) = \sup_{\Pi \in \mathcal{P}_B(I)} G_B(\Pi; I).$$

Protocol loss is $L_B(\Pi; I) = V_B(I) - G_B(\Pi; I)$, hence

$$G_B(\Pi; I) = V_B(I) - L_B(\Pi; I).$$

This is an accounting identity (DD-C-0002), not an efficiency theorem. Its use is to prevent a comparison that changes information, authority, equilibrium selection, and source dependence simultaneously from being assigned to only one cause.

Frontier monotonicity requires emulation. If every protocol feasible under I_1 can be implemented under I_2 with the same outcome distribution, then $V_B(I_2) \geq V_B(I_1)$ (DD-C-0016). The condition is stronger than saying that one realized protocol sees more information. A richer public signal can change a game and its selected outcome even when the planner frontier rises. This distinction is the hinge between the disclosure result and Blackwell's decision-theoretic order [Blackwell, 1953, Kamenica and Gentzkow, 2011].

2.2 Three institutional margins

The results occupy different cells of an information–authority matrix. A private team chooses role policies ex ante, observes only local signals at action time, and shares a common discovery objective. A disclosure game gives all agents a public posterior but leaves actions strategic under an equal-split reward. A source network changes how private reports are jointly generated while holding the direct action rule fixed. These are not three algorithms for the same feasible set; each changes a different primitive.

For the private team, ex-ante randomization cannot improve the finite common-payoff optimum (DD-C-0018). Drawing contingent actions before signals produces a mixture over deterministic profiles, and expected discovery is a convex combination of their values. This is a team-decision sufficiency result, not purification of a strategic mixed equilibrium [Marschak and Radner, 1972].

A pooled planner provides an important but carefully scoped comparator. It observes every private report and can apply any fixed role policy to reproduce the team's action vector. Conditional on the reports, an eight-label top-posterior portfolio weakly dominates any emulated multiset. Thus direct clue-following is a lower bound and the pooled frontier is an upper bound under matched physical actions and objective (DD-C-0019). The comparison crosses information and assignment margins; it is a valid feasible-set bound, not an attainable private policy.

2.3 Claim-level evidence

The numerical and mathematical objects below do not share one evidence status. A theorem can be derived analytically and audited computationally. A finite enumeration can be exact without supporting an all-parameter theorem. A selected equilibrium reversal can be verified while remaining selection-dependent. A failed exhaustive search over a declared finite class is a bounded null, not a general sufficiency statement. Table 1 makes these differences visible.

Table 1: Evidence status is attached to each result, not to the paper as a whole.

Object	Evidence	Status	Boundary
Policy-signature theorem	DD-C-0023 theorem	derived	Analytic proof; computation audits implementation.
Hybrid witness $7/10 > 16/25$	DD-C-0021 bounded exact	independently reproduced	One finite fixture; refutes general direct optimality.
Disclosure reversal	DD-C-0030 selected exact	verified	Selection-dependent; pure equilibria and planner improve.
Selection robustness census	DD-C-0040 bounded exact	independently reproduced	Known reversal survives only anonymous-symmetric selection among six audited rules.
Heterogeneous pairwise counterexample	DD-C-0043 bounded exact	independently reproduced	Exact colored class; not an arbitrary-source theorem.
Canonical private-team optimum	DD-C-0038 exact optimum	independently reproduced	Frozen zero-communication fixture; not an all-parameter theorem.

Source: `claims/claims.yml`; claims: DD-C-0021, DD-C-0023, DD-C-0030, DD-C-0040, DD-C-0043, DD-C-0038; generator: `distributed_discovery.papers.build_three_results`; input SHA-256 prefix: `2241903048b5`.

Every computational statement traces to a passing immutable run with a frozen configuration, clean source commit, environment, exact or floating arithmetic basis, termination condition, output checksums, and validation flags. Independent reproduction requires a materially different representation or evaluator, not a second invocation of the same code. Corruption tests establish that certificate checkers actually reject altered evidence. These practices do not replace proof or peer review; they make the boundary of each claim inspectable.

3 Private roles without communication

3.1 The zero-communication team problem

The target is uniform on M labels. Agent i receives clue X_i with

$$\mathbb{P}(X_i = \theta \mid \theta) = p, \qquad \mathbb{P}(X_i = x \neq \theta \mid \theta) = q = \frac{1-p}{M-1},$$

independently conditional on θ. Before clues are realized, the team commits to deterministic role policies $f_i : \{1,\ldots,M\} \to \{1,\ldots,M\}$. Afterward agent i searches $f_i(X_i)$ without communication. The team frontier is

$$T_N(M,p) = \max_{f_1,\ldots,f_N} \mathbb{P}\{\theta \in \{f_1(X_1),\ldots,f_N(X_N)\}\}.$$

Direct clue-following uses $f_i(x) = x$. Its value is $1-(1-p)^N$. A territorial policy ignores clues and reserves a fixed label. Hybrid policies combine both: one role can protect territory while another redirects only selected signals. The objective rewards coverage, so sacrificing some action-level accuracy can improve union success.

For a fixed profile, let $s_i(t) = \mathbb{P}(f_i(X_i) = t \mid \theta = t)$. Conditional independence yields the exact factorization

$$G(f_1,\ldots,f_N) = \frac{1}{M}\sum_{t=1}^{M}\left[1 - \prod_{i=1}^{N}(1 - s_i(t))\right].$$

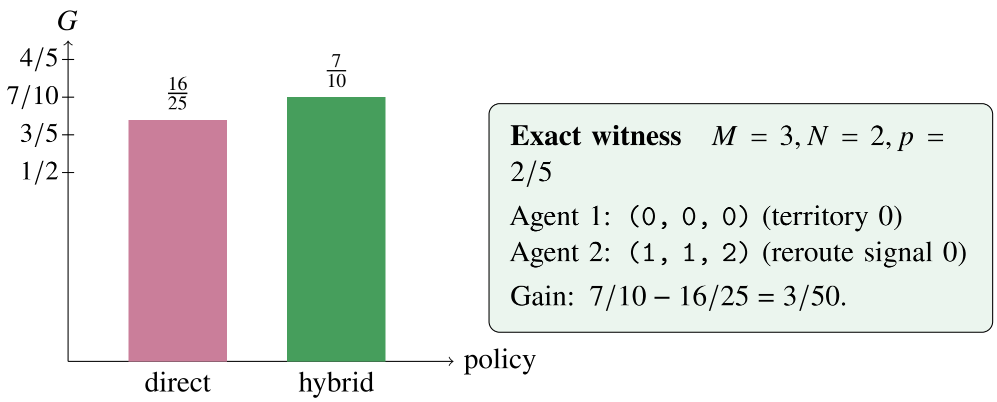

Figure 1: Roles can improve discovery without communication. The bars use the exact unrestricted optimum at the displayed finite fixture; the schematic is one optimal representative.

Source runs: `20260720T200447Z_DD-001_6eb12861_ba766d1eba`, `20260720T223829Z_DD-001_b2cc23f4_5e16a90ad1`; claims: DD-C-0021, DD-C-0026, DD-C-0027, DD-C-0028; generator: `distributed_discovery.papers.build_three_results`; input SHA-256 prefixes: `e7112e7dd471, 47d933bfc1ae`.

This evaluator is separate from the direct target/signal enumeration used to check winners in the finite grid.

3.2 An exact role counterexample

At $M = 3$, $N = 2$, and $p = 2/5$, direct clue-following gives

$$D(p) = 1 - (1 - p)^2 = 2p - p^2 = \frac{16}{25}.$$

The hybrid profile with one constant policy $(0, 0, 0)$ and one rerouting policy $(1, 1, 2)$ gives $7/10$. Exhaustive enumeration of all agent-symmetric policy profiles proves this is the unrestricted deterministic optimum for the fixture; raw direct enumeration and a later signature implementation reproduce it (DD-C-0021, DD-C-0024). The exact gain is $3/50$.

The counterexample has a transparent mechanism. The constant role guarantees an attempt at label zero. The other role follows signals one and two but maps a zero signal to one, avoiding predictable duplication with the territorial agent. It does not show that one fixed territory is always best, that communication is harmful, or that the same profile is optimal at canonical dimensions. It refutes only the general claim that every optimal private team follows its clue directly.

3.3 A restricted phase theorem

For $N = 2$ and $M \geq 3$, compare three families: two direct policies; two distinct constant territories; and the one-reroute hybrid above. Their values are

$$D(p) = 2p - p^2, \qquad R_M = \frac{2}{M}, \qquad H_M(p) = \frac{1 + (M - 2)p}{M - 1}.$$

Exact factorization of pairwise differences gives the thresholds $p = 1/M$ and $p = 1/(M - 1)$ (DD-C-0026). Territorial is best below $1/M$, the hybrid is best between the thresholds, and direct is best above $1/(M - 1)$, with adjacent ties at the boundaries. This is a theorem within the three named families for every $M \geq 3$.

The restriction matters. A complete signature-polynomial census certifies that the same hybrid-then-direct envelope is unrestricted over the continuous informative interval $[1/M, 1]$ for $M = 3, 4, 5$ (DD-C-0027). The computation checks 438,734 feasible signature multisets with exact quadratic nonnegativity at endpoints and interior vertices. It does not prove the envelope for all M.

Nor does the informative classification extend to every p. At $p = 0$, exact unrestricted optima are $11/12$ for $M = 3$ and $2/3$ for $M = 4$, strictly above the three-family benchmarks $2/3$ and $1/2$ (DD-C-0028). These anti-informative counterexamples show why a phase diagram needs an explicit parameter domain.

3.4 Lossless policy signatures

Raw policies number M^M per agent. A policy can nevertheless be compressed without changing any fixed-profile value. For target t, let $c_i(t)$ be the number of signals mapped to t, and let $d_i(t)$ indicate whether signal t is a fixed point. Then

$$s_i(t) = qc_i(t) + (p - q)d_i(t).$$

The targetwise pairs $(c_i(t), d_i(t))$ therefore determine the full conditional success vector and every profile objective (DD-C-0023).

Feasibility also has an exact characterization. Remove fixed rows and define residual incoming counts $r_t = c_t - d_t$. A duplicated-column bipartite matching exists exactly when local count conditions and the singleton residual Hall bounds hold. A matching implementation reconstructs policies, while a separate closed- form implementation audits feasibility. The signature representation independently reproduces all 21 prior tiny optima and their raw-policy tie counts (DD-C-0024).

The compression does not eliminate the canonical difficulty. At $M = 16$ there are 148,348,284,928 feasible labeled signatures and 5,806 individual target-label orbits. Eight-agent multisets before a global target quotient form an 85-digit state count (DD-C-0025). Canonicalizing each agent separately loses relative target alignment, which enters the product of failure probabilities. The barrier is not a proof that exact optimization is impossible; it identifies what a valid next quotient or relaxation must preserve.

3.5 The exact canonical optimum

For $(M, N, p) = (16, 8, 1/5)$, direct clue-following is feasible and has exact value

$$1 - (4/5)^8 = \frac{325089}{390625} = 0.83222784.$$

The direct profile is an exact coordinate fixed point, and 17 additional declared starts converged to the same value (DD-C-0022). Those searches alone were only heuristic lower-bound evidence. The new proof instead retains the aligned vector of incoming policy counts across all agents inside each target column. It lower-bounds each local failure factor, exactly minimizes their product by column resource, and then relaxes only cross-target feasibility while preserving the globally necessary incoming-count total (DD-C-0037).

The resulting exact Bellman certificate assigns resource eight to each of the sixteen targets and lower-bounds average failure by $(4/5)^8$. Hence

$$\boxed{T_8(16, 1/5) = \frac{325089}{390625} = 0.83222784.}$$

The upper bound meets the attainable direct value, proving global optimality for deterministic teams. An ex-

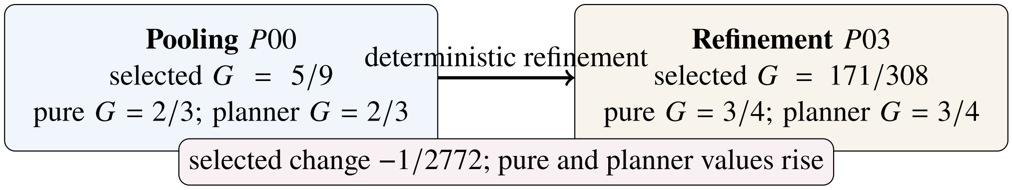

Figure 2: The bounded disclosure reversal is selection-dependent. The selected anonymous-symmetric outcome falls, whereas every pure-equilibrium value and the two-action planner value rise for the same witness.
Source runs: 20260720T225848Z_DD-002_94607423_e29b1460ae; claims: DD-C-0029, DD-C-0030, DD-C-0031; generator: distributed_discovery.papers.build_three_results; input SHA-256 prefixes: ff32fc8cb5cb.

ante randomized policy is a mixture of deterministic profiles, so the randomized optimum is identical (DD-C-0038). A separate verifier checks every Bellman inequality and equality witness without optimizing and rejects a corrupted final value. The former pooled upper endpoint remains a valid organizational benchmark, but its exact gap has now been closed for this frozen private-team problem.

4 Information disclosure under strategic search

4.1 A complete bounded fixture

The disclosure result holds information and the physical action technology apart from strategic selection. There are three target actions, two searchers, four evidence states, and a fixed rational likelihood matrix. A deterministic public disclosure policy is a set partition of the four evidence states, considered up to message-label equivalence. The Bell number gives exactly 15 policies. Across their messages the run evaluates 37 posterior games (DD-C-0029).

Agents compete for an equal-split prize: being the sole correct searcher receives the whole prize, while duplicate correct searches split it. For each posterior game, the computation exhausts labeled pure best responses and calculates the anonymous symmetric equilibrium over its support. Cartesian products across messages yield 256 global pure-equilibrium selections. This registry keeps the correspondence separate from the named symmetric selection.

A disclosure π' is more informative than π when a deterministic garbling maps messages of π' to those of π. The complete partition lattice contains 45 strict refinement pairs. The planner's two-action value is the posterior mass of its two highest-probability targets and therefore cannot fall under a refinement when it can ignore the added message detail.

4.2 The selected reversal

The unique anonymous-symmetric reversal refines full pooling $\{\{0, 1, 2, 3\}\}$ into $\{\{0, 1\}, \{2, 3\}\}$. Under pooling the selected discovery is $5/9$. Under refinement it is the probability-weighted value $171/308$, lower by $1/2772$ (DD-C-0030).

The qualification is central. Every pure-equilibrium discovery value rises from $2/3$ to $3/4$ for this witness. The two-action planner value also rises from $2/3$ to $3/4$. The refinement therefore does not make every equilibrium worse, does not reduce the feasible information frontier, and does not show that public information is intrinsically harmful. It changes the declared anonymous-symmetric outcome in a direction that concentrates strategic search.

The witness is informational-Braess-like because information modifies equilibrium behavior, while a coordinator could discard the added detail or choose a better portfolio [Acemoglu et al., 2018]. It is also narrower than a Bayesian-persuasion claim: the disclosure policy is not optimized over randomized experiments, the sender objective is not solved, and the result does not characterize the concave closure of value [Kamenica and Gentzkow, 2011].

4.3 What the full refinement census says

Across all 45 strict deterministic refinements, exactly one lowers the selected anonymous-symmetric value, eight lower the worst pure-equilibrium value, none lower the best pure-equilibrium value, and none lower the planner value (DD-C-0031). Reporting only the selected witness would hide the favorable pure correspondence; reporting only the best equilibrium would hide selection risk. Both summaries are needed.

4.4 Exact selection catalogue

The same finite fixture admits an exact-potential characterization. For posterior target probabilities μ, define

$$\Phi(a,b) = \begin{cases} 3\mu_a/2, & a = b, \\ \mu_a + \mu_b, & a \neq b. \end{cases}$$

Every unilateral payoff difference equals the corresponding potential difference, so every strict best-response path terminates (DD-C-0039). This permits an exact catalogue beyond the initially declared anonymous-symmetric rule: best and worst pure equilibria, uniform selection over global potential maximizers, and a process that starts uniformly over the nine labeled profiles and chooses uniformly among strict payoff-maximizing player/action moves. The planner remains a nonstrategic comparator.

Table 2: Selection robustness of the known witness and the complete 45-refinement census.

Rule	$G(P00)$	$G(P03)$	Reversal?	Harmful refinements
Anonymous symmetric	$5/9$	$171/308$	yes	1
Best pure	$2/3$	$3/4$	no	0
Worst pure	$2/3$	$3/4$	no	8
Uniform potential maximum	$2/3$	$3/4$	no	2
Uniform strict-BR basin	$2/3$	$3/4$	no	2
Planner	$2/3$	$3/4$	no	0

Potential-maximum and basin values coincide on all 15 policies; five policies nevertheless expose multiple potential discovery values and branch-dependent best-response basins.

Source runs: `20260721T025802Z_DD-002_73a85c71_b0e5b6dc49`; claims: DD-C-0039, DD-C-0040, DD-C-0041; generator: `distributed_discovery.papers.build_three_results`; input SHA-256 prefixes: `657474921dcf`, `2325dc263c53`.

For the known $P00 \to P03$ witness, best pure, worst pure, potential-maximum, best-response-basin, and planner values all rise from $2/3$ to $3/4$; only the anonymous-symmetric value falls (DD-C-0040). Across all 45 refinements, the harmful counts in table order are $1, 0, 8, 2, 2, 0$. Thus selection risk is not confined to the original rule—worst pure and the two dynamic/potential catalogues have other harmful refinements—but the named witness itself is not robust across the five alternatives (DD-C-0041). An independent implementation checks the posterior games, potential identities, absorption Bellman equations, and all rule comparisons.

The census establishes completeness only for the declared four-state partition lattice and likelihood fixture. Randomized disclosure was not studied. General asymmetric mixed equilibria were not enumerated. Dynamic learning, equilibrium refinement, and selection processes outside the audited strict-best-response rule are outside the result. The catalogue resolves the planned robustness milestone; expanding to randomized disclosure requires a separate solution concept and ADR.

5 Correlated source networks

5.1 From a copying scalar to provenance graphs

Four searchers observe reports derived from one to three latent sources. A bipartite graph says which source reports each searcher sees; no declared node is isolated. Sources are independent conditional on a three-valued target and have common accuracy $2/3$. A searcher combines adjacent source reports by a declared plurality-winner-set rule with uniform tie sampling, then searches the resulting label directly.

This model turns provenance into a graph rather than a single copying rate. Source and searcher labels are nuisance symmetries, so graphs are equivalent under independent relabeling of the two sides. Exhausting all labeled adjacencies and canonicalizing them yields 1, 8, and 42 nonisomorphic graphs for one, two, and three sources: 51 total (DD-C-0032). An independent orbit traversal reproduces the counts and a pairwise nonisomorphism audit confirms the registry.

For each graph, exact enumeration over target and source-signal states produces private discovery and report moments. The complete pairwise object includes every searcher pair's one-hot report moment matrix, canonicalized only under searcher relabeling. Two networks can have the same scalar mean agreement while differing in this matrix, source overlap, and higher-order dependence.

5.2 A bounded null for complete pairwise moments

The 51-graph census contains ten complete pairwise-moment signatures that each match two nonisomorphic graphs. All twenty graphs in those groups have equal private discovery within their matched group. Thus no complete-pairwise-moment counterexample exists in the configured class (DD-C-0033).

This is a bounded null, not a sufficiency theorem. It says the declared diagnostic survived one complete exact test. A counterexample may appear with heterogeneous source quality, more sources, different aggregation, more searchers, or another target law. Conversely, the null matters operationally because it prevents a failed search from being selectively forgotten and establishes a reproducible baseline for the next perturbation.

5.3 Heterogeneous accuracy breaks the null

The registered perturbation carries each source accuracy as a color attached to its adjacency row. Searchers may still be relabeled, while source relabeling moves the exact color with the source. Four base profiles—equal $2/3$, modest $\{1/2, 2/3\}$, near-uninformative $\{1/3, 2/3\}$, and strong $\{1/3, 5/6\}$—produce 41,612 valid labeled colored objects and 671 orbits. One controlled three-color expansion adds 12,966 labeled objects and 168 orbits (DD-C-0042). The equal profile reproduces all prior two/three-source values.

The complete-moment null fails already in the modest two-source profile. One network attaches the $1/2$ source to one searcher and the $2/3$ source to all four; the other reverses those degrees. They are not color-

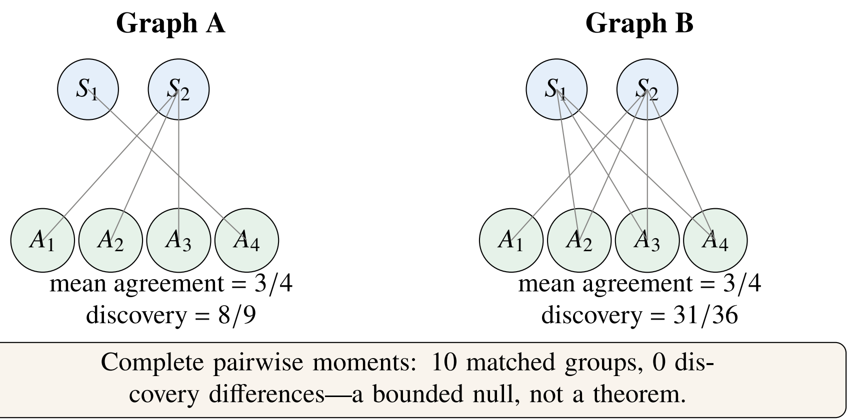

Figure 3: Two nonisomorphic source networks have equal mean pair agreement but different exact private discovery. Their complete pairwise moment matrices differ.
Source runs: `20260720T232223Z_DD-003_2ea8dad5_ae62f6c1f1`; claims: DD-C-0032, DD-C-0033, DD-C-0034; generator: `distributed_discovery.papers.build_three_results`; input SHA-256 prefixes: `92132b3bb641, 1c8675d8697a`.

preserving isomorphic. Every first action moment and all six 3×3 pairwise action-moment matrices match, but private union discovery is 3/4 versus 2/3 (DD-C-0043).

Table 3: A heterogeneous-source counterexample with identical complete first and pairwise scalar-report moments.

Network	Degree of 1/2 source	Degree of 2/3 source	Discovery
A	1	4	3/4
B	4	1	2/3

Both have pair-agreement signature (17/24, 17/24, 17/24, 1, 1, 1) and the same full 66-entry moment signature. Across 839 colored networks, 163 signatures match multiple networks and 111 matched groups differ in discovery.
Source runs: `20260721T032358Z_DD-003_84238b76_2cbc13e66a`; claims: DD-C-0042, DD-C-0043, DD-C-0044; generator: `distributed_discovery.papers.build_three_results`; input SHA-256 prefixes: `8bd4c3bc7f81, 8f8f28251887`.

Across the full 839-network registry, 163 complete-moment signatures match 485 networks and 111 matched groups contain different discovery values (DD-C-0044). A separate implementation traverses the labeled colored orbits, directly checks nonisomorphism, and reconstructs every 66-entry signature and discovery value. Thus heterogeneity supplies an exact bounded counterexample, not a universal impossibility theorem for all pairwise diagnostics.

5.4 A scalar counterexample

Average pair agreement is much weaker. Graphs `K2N4:00011110` and `K2N4:01111111` both have mean pair agreement 3/4, yet their exact private discovery probabilities are 8/9 and 31/36, a difference of 1/36 (DD-C-0034).

The two complete pairwise matrices differ, so the example refutes only the scalar average. The full census has 22 graph pairs with matched average agreement and different discovery. It also has 59 pairs with matched source Herfindahl– Hirschman concentration and different discovery. Neither scalar is a sufficient effective-channel statistic in this model.

The result does not imply that correlation is always harmful. Shared sources may improve individual report quality or coordinate complementary actions in another technology. Nor does it establish a universal effective sample size. Diagnostics must be tied to the target law, aggregation rule, and action objective. Provenance structure can matter beyond nominal report count, but which representation is sufficient remains a model-specific question.

6 Unified implications

The three results share an outcome—union discovery—but operate through different causal and mathematical margins. It is useful to state the mapping exactly:

Study	Primitive changed	Mathematical object	Main boundary
DD-001 roles	ex-ante policy assignment	common-payoff team frontier	canonical optimum exact; other parameters open
DD-002 disclosure	public signal partition	posterior game plus selection	known reversal survives one of six audited rules
DD-003 sources	joint evidence provenance	colored/bipartite source law	heterogeneity breaks the homogeneous pairwise null

First, allocation can substitute partly for communication. The DD-001 hybrid improves coverage using differentiated role mappings while leaving every clue private. The result does not say that information sharing lacks value; the exact pooled planner remains an upper bound. It says assignment has an independent margin that a direct symmetric protocol leaves unused.

Second, information and behavior must be separated. The DD-002 refinement improves the planner and all pure equilibria for its witness while lowering one selected symmetric outcome. Five audited alternatives do not reproduce that witness, although other refinements are harmful under worst-pure, potential, and best-response-basin summaries. A statement that "information reduced discovery" is incomplete without the selection procedure. Frontier monotonicity and realized equilibrium performance can move in different directions.

Third, evidence diversity needs a representation. The homogeneous DD-003 census preserves a bounded null for the full pairwise object, while exact source-accuracy colors break it: pairwise observational equivalence can coexist with different union discovery. A useful diagnostic must be validated against the downstream action rule and perturbations of the generative law, not chosen solely because it resembles a generic effective sample size.

These points support a layered design workflow. Declare the world and evidence kernel; declare what each agent observes; declare the feasible role or assignment class; declare rewards and equilibrium selection; declare how actions cover the target space. Only then compare protocols. A change that bundles those layers may still be useful, but its performance gap should not be given a single causal label.

The scientific-search literature already studies richer dynamic divisions of labor, communication networks, and experiment choice [March, 1991, Kitcher, 1990, Zollman, 2007, Rzhetsky et al., 2015]. Submodular and adaptive search add overlapping value and feedback [Nemhauser et al., 1978, Golovin and Krause, 2011]. Robotic coverage makes redundancy valuable for robustness [Hazon and Kaminka, 2008]. The present finite models are best read as diagnostic building blocks for those settings, not substitutes for them.

7 Limitations

The first limitation is external validity. Every result is mathematical evidence inside a declared model. No observational or experimental dataset establishes that a real scientific group, company, market, or multi-agent system occupies the same parameterization. Application mappings identify measurable primitives; they do not transfer the ordering automatically.

Second, the role results are asymmetric in their strength. The policy-signature identity and residual Hall characterization are general theorems for the finite symmetric clue model. The three-family phase result is general only within its named families. The unrestricted continuous classification is bounded to $M = 3, 4, 5$, and the anti-informative examples are two fixtures. The alignment relaxation upper-bounds all symmetric-report parameters, but is not always tight; its equality with direct clue-following proves the optimum only at the frozen canonical fixture.

Third, the disclosure result fixes deterministic partitions of four evidence states and six specified evaluations. It does not enumerate randomized disclosure, general asymmetric mixed equilibria, or learning processes beyond the audited strict-best-response basin. Because the known witness reverses under only one of the six evaluations, the phrase "harmful disclosure" must always carry the selection qualifier.

Fourth, the source-network model fixes source independence conditional on the target, four searchers, two or three sources for the colored census, a finite accuracy palette, and a particular plurality/tie rule. The complete pairwise counterexample is exact only there. It does not cover continuous accuracies, dependent sources, asymmetric target priors, larger graphs, or higher-order report laws.

Fifth, exact enumeration is not equivalent to conceptual independence. The repository requires distinct representations and direct state checks where feasible, but all implementations share the frozen model. A misspecified likelihood, objective, or selection definition could be reproduced perfectly. Proof review and model criticism remain necessary.

Finally, the project makes no broad terminology or field novelty claim (DD-C-0017). Team theory, information design, organizational search, epistemic networks, and coverage games provide direct intellectual ancestry. The value of the synthesis is an inspectable separation of information, roles, selection, and source representation.

8 Research agenda

The immediate DD-001 proof task is complete: the first alignment-preserving count-budget relaxation equals the direct lower bound canonically. The next structural question is when that relaxation is tight. It matches all 21 prior tiny fixtures but is strictly loose at the audited anti-informative $(M, N, p) = (3, 2, 0)$ case. Stronger per-role cross-target consistency can be added as a hierarchy only when a new question requires it.

For DD-002, the selection-robustness milestone is complete. Six fully specified evaluations now classify the known witness and all 45 refinements, with exact best-response absorption witnesses. A future randomized-disclosure study should begin only after a separate ADR defines the experiment class, sender objective, and solution object; it is not implied by the deterministic catalogue.

For DD-003, the registered accuracy perturbation is complete. The two-level base and single three-color expansion establish exact full-pairwise counterexamples. Any next step should pose a new bounded identification question—for example one asymmetric prior or dependent-source law—and audit its state size before expanding; no unbounded graph search follows automatically.

Longer-term work introduces feedback, overlapping coverage, mechanism design, and empirical audits.

Sequential search requires a stopping and information-update model; greedy guarantees require submodularity or adaptive submodularity rather than analogy [Nemhauser et al., 1978, Golovin and Krause, 2011]. Mechanism design must specify observability, budget balance, and credit. Empirical work must distinguish descriptive concentration from an identified frontier or protocol-loss estimate.

Across these extensions, the evidence policy should remain constant: preregister the feasible class and stopping rule, estimate the state space before computation, preserve failed and null results, use immutable run identifiers, and promote claims only after every proof, code, run, and test path has been audited.

9 Conclusion

Three finite results expose three distinct ways a collective search can depart from a symmetric direct protocol. Roles can reserve coverage without communication; public disclosure can change a selected strategic outcome even as the planner and pure equilibria improve, while alternative selection rules can reject the same reversal; source networks can share complete first and pairwise moments while producing different discovery once accuracy is represented as a color.

The synthesis is deliberately qualified. The role counterexample is exact but finite. The disclosure reversal is exact, selection-dependent, and not robust to the other five audited evaluations. The homogeneous pairwise source result remains a bounded null, while the heterogeneous colored class supplies an exact bounded counterexample. The canonical private-team optimum is exact, but only for the frozen zero-communication dimensions and symmetric signal law.

The shared lesson is architectural rather than universal: collective discovery depends not only on what agents know, but on how roles are assigned, how strategic outcomes are selected, and how source dependence is represented. Making those margins explicit turns an appealing paradox into a sequence of falsifiable, auditable research problems.

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